

Unit A1: Computer Fundamentals (Core)

Sub-topics: A1.1 to A1.3 (SL Content)

1. Syllabus Scope & Learning Outcomes

The "Do" column defines the practical skills required for the exam (Paper 1).

Syllabus Point	Command Term	Student Expectation (The "Do")
A1.1.1	Describe	Describe the function of the ALU, CU, Registers (PC, MAR, MDR, ACC, IR), and Buses (Data, Address, Control).
A1.1.2	Describe	Describe the role of a GPU (specialized circuit for parallel processing of images/video/AI).
A1.1.4	Explain	Explain the purposes of primary memory types: RAM (Volatile), ROM (Non-volatile), and Cache (L1/L2/L3).
A1.1.5	Describe	Trace the Fetch-Decode-Execute cycle step-by-step (e.g., using Little Man Computer logic).
A1.1.7	Describe	Describe secondary storage types: Internal (HDD/SSD), External (Optical, Flash), and NAS.
A1.1.8	Describe	Describe compression concepts: Lossless (RLE) vs. Lossy (JPEG/MP3).

A1.1.9	Describe	Describe cloud services: SaaS (Software), PaaS (Platform), IaaS (Infrastructure).
A1.2.1	Describe	Convert integers between Binary (Base-2), Decimal (Base-10), and Hexadecimal (Base-16).
A1.2.2	Explain	Explain how binary stores data: Characters (ASCII/Unicode), Images (Bitmaps), Sound (Sampling).
A1.2.3	Describe	Identify logic gates: AND, OR, NOT, NAND, NOR, XOR, XNOR.
A1.2.4	Construct / Analyse	Create truth tables for logic circuits with up to 3 inputs.
A1.2.5	Construct	Draw logic circuit diagrams from Boolean expressions (e.g., $Y = (A \cdot \quad)$).
A1.3.1	Describe	Describe the role of an OS: Managing resources (CPU, Memory, Peripherals) and providing a UI.
A1.3.2	Describe	Describe OS functions: Memory Management, File Management, Device Management (Drivers), Security.
A1.3.3	Compare	Compare scheduling algorithms: FCFS (First Come First Served), Round Robin, Priority.

A1.3.4	Evaluate	Evaluate Interrupts vs. Polling for handling peripheral events.
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2. Key Terminology & Definitions

- **CPU (Central Processing Unit):** The "brain" of the computer, executing instructions.
- **ALU (Arithmetic Logic Unit):** Performs math (+, -, *, /) and logic (AND, OR) operations.
- **CU (Control Unit):** Directs operations, managing the Fetch-Decode-Execute cycle.
- **Registers:** Small, fast storage inside the CPU (e.g., **Program Counter** holds the address of the next instruction).
- **Buses:**
 - **Data Bus:** Carries the actual data (Bidirectional).
 - **Address Bus:** Carries the location address (Unidirectional).
 - **Control Bus:** Carries command signals (Read/Write).
- **Cache:** Fast memory between CPU and RAM. Levels: L1 (Fastest/Smallest) -> L3 (Slowest/Largest).
- **Volatile:** Data is lost when power is off (RAM). **Non-Volatile:** Data remains (ROM, HDD).
- **Fetch-Decode-Execute:** The cycle of retrieving an instruction, working out what it is, and carrying it out.
- **Logic Gate:** A building block of digital circuits.
- **Truth Table:** A table showing all possible input combinations and the resulting output.
- **Operating System (OS):** Software managing hardware and providing services for programs.
- **Interrupt:** A signal stopping the CPU to handle an urgent task (e.g., mouse click).
- **Polling:** The CPU regularly checking if a device needs attention (inefficient).

3. Core Content & Exam Actions

A1.1: Hardware & Operation

Theory:

- **Von Neumann Architecture:** Shared memory for data and instructions.
- **Storage:**
 - **HDD:** Magnetic platters, moving parts, cheaper/GB.
 - **SSD:** Flash memory, no moving parts, faster, durable.
- **Compression:**
 - **Lossless (RLE):** AAAAABBB -> 5A3B. Original data perfectly restored. Used for text/code.
 - **Lossy (JPEG):** Discards "invisible" data to save space. Cannot restore original. Used for media.

Student Exam Action:

- **Tracing:** Describe the exact steps of the Fetch phase: PC -> MAR -> Address Bus -> RAM -> Data Bus -> MDR -> CIR -> PC+1.

A1.2: Data Representation

Theory:

- **Binary:** Base-2 (0, 1).
- **Hexadecimal:** Base-16 (0-9, A-F). Used to shorten binary for humans (e.g., color codes #FF0000).
- **ASCII vs Unicode:** ASCII is 7-bit (128 chars, English only). Unicode is 16/32-bit (supports all global languages/emojis).
- **Images:**
 - **Bitmap:** Grid of pixels. File size = $Resolution \times ColourDepth$.
 - **Colour Depth:** Bits per pixel (e.g., 8-bit = 256 colors).
- **Sound:** Analogue waves are **Sampled** at intervals.
 $SampleRate(Hz) \times BitDepth \times Duration$.

Student Exam Action:

- **Calculation:** Convert 1101 (Binary) to Decimal (13) and Hex (D).
- **Logic:** Given the expression $X = A \text{ AND } (B \text{ OR } C)$, complete the Truth Table.

A1.3: Operating Systems

Theory:

- **Memory Management:** Allocating RAM to programs. Using **Virtual Memory** (HDD acting as RAM) when RAM is full (slower due to "paging").
- **Scheduling:**
 - **FCFS:** Simple queue. Can cause "convoy effect" (slow process blocking fast ones).
 - **Round Robin:** Time slices (Quantum). Fair, but context switching has overhead.
- **Device Drivers:** Translators between the OS and specific hardware.

Student Exam Action:

- **Evaluation:** Discuss why a Real-Time OS (e.g., car airbag) uses Interrupts instead of Polling (Need immediate response vs wasting CPU cycles checking).

4. Common Pitfalls & Examiner Tips

- **Pitfall (Buses):** Students often forget that the **Address Bus** is usually unidirectional (CPU -> Memory), while Data/Control buses are bidirectional.
- **Pitfall (Logic Gates):** Confusing **NAND** (Not AND) with **NOR** (Not OR). Remember: NAND is 0 only when *both* inputs are 1.

- **Tip (Conversions):** When converting Hex 1F to Decimal, do $1 \times 16 + 15 \times 1 = 31$. Don't guess.
- **Tip (OS):** Don't just list "Windows" as an OS. Define *what it does* (Interface, Resource Management).